Probability and Statistics B - Practical Sheet 1

Getting started:

1. Open **RStudio** (do not use the base R application).
2. From the menu, select **File** → **New File** → **R Script**. This will create a blank script where you can type and run your R commands.
3. Immediately save your script with a meaningful name, e.g., `Practical1.R`, using **File** → **Save As**. Choose a location you can easily find later.
4. Type your commands in the script and run them using:
   - **Ctrl + Enter** (Windows) or
   - **Cmd + Enter** (Mac)
5. When you have your answers, copy the relevant output (not the entire script) into a Word document.

Important: Always keep your R script saved. This is good practice and ensures you have a record of your work.
If you are unsure about how any command works, use, e.g., `?hist`.

1. Enter in the data

   ```
   60 85 72 59 37 75 93 7 98 63 41 90 5 17 97
   ```

   Use any method you like, e.g.

   ```
   sample1 = c(60, 85, 72, 59, 37, 75, 93, 7, 98, 63, 41, 90, 5, 17,
   97)
   ```

   Make a stem and leaf plot. (The command is "stem")

2. Read this stem and leaf plot, enter in the data and make a histogram:

   The decimal point is 1 digit(s) to the right of the |

   ```
   8 | 028
   9 | 115578
   10 | 1669
   11 | 01
   ```

   (The command is "hist")

3. One can generate random data with the "R" commands. For example

   ```
   x = rnorm(100)
   ```

   produces 100 random numbers with a standardised normal distribution. Create two different histograms for two different times of defining x as above. Do you get the same histogram?

4. Data were collected on the time (in days) until each of 200 claims is settled by the insurer in a certain insurance portfolio. The data are given in the file `settle_time.txt` which can be found on Canvas in the week 1 module.

(a) Download the data on your computer and read the values into R (a useful command in R is `settle.times=scan()`).

(b) Summarise the data using the mean and the five point summary, and comment on the symmetry of the distribution of the data based on these measures (useful commands in R are `mean`, `quantile()`, `summary()`).

(c) Produce a frequency table of the times to settlement. It may be useful to divide the range of the data into intervals of 30 days, i.e. [0, 30), [30, 60), ... [390, 420). Note that whether an interval is right-open or left-open will make a difference. In R you can do this as follows:

```
settle.breaks = seq(0,420,by=30)
settle.table = table(cut(settle.times, breaks = settle.breaks,
right=F))
settle.table
```

(d) Draw a histogram of the data (using the same intervals as in the table above) and a boxplot. Useful commands in R are:

```
hist(settle.times,breaks=settle.breaks,freq=TRUE,col="gray",
main="Histogram of claims settling time",xlab="Days")
boxplot(settle.times, horizontal = T)
```

(e) Comment briefly on the shape of the distribution of the data. Can you suggest a suitable theoretical distribution for the time until claims are settled in this portfolio?

5. The following data are a sample from a survey on property prices (in £1000) in a city in Scotland:

```
191.6 216.7 196.6 220.7 301.0 236.1 142.4 190.4 200.9 241.8
184.9 202.1 156.7 198.1 240.1 169.7 151.8 257.0 347.9 281.6
```

In addition to the data above, suppose that a similar survey in a different Scottish city gave the following property prices (again in £1000):

```
130.4 155.1 155.7 187.2 100.4 142.8 118.3 154.8 121.1 126.1
137.8 158.7 147.0 132.3 133.0 127.0 88.6 98.8 155.4 124.9
```

(a) Compare the dispersion of the two data sets using histograms (remember to put the two data sets on the same scale for comparison).

(b) Compare the dispersion of the two data sets using boxplots again putting the two data sets on the same scale for comparison.

(c) Now compare the dispersion of the two samples using appropriate numerical measures of spread. (Hint: is the variance suitable here?)

6. Produce functions in R to calculate the coefficient of symmetry and the kurtosis using the code below.

```
beta1.fn = function(x)
{
        sum.dev3 = sum( (x - mean(x))^3 )
        b1 = (sum.dev3/length(x))/(sd(x))^3
        cat("\n","Coefficient of symmetry for these data is: ",
        b1, "\n\n")
}

kurtosis.fn = function(x)
{
        sum.dev4 = sum( (x - mean(x))^4 )
        k1 = ((sum.dev4/length(x))/(sd(x))^4)-3
        cat("\n","Kurtosis for these data is: ", k1, "\n\n")
}
```

Find the mean, standard deviation, coefficient of symmetry and kurtosis for the following data:

```
33 23 34 27 32 45 5 50 27 25 25 8 27 13 31 9 16 10 40 35 41 4 22
41 43 52 25 40 27 22
```

7. Download a new library to your R package using the command:

```
install.packages("UsingR")
```

Choose any mirror site you like (St. Andrews worked for me). Obtain and attach the data set `cfb` by typing the following:

```
library(UsingR)
data(cfb)
attach(cfb)
```

One of the vectors in this data frame is called `INCOME`. It represents the yearly income figures by households in a survey of consumer finances in the US in 2001. You can see more information about it by using

```
?cfb
```

   (a) Obtain a histogram of the `INCOME` data. What does it tell you?

   (b) Calculate the mean income using `mean(INCOME)`. How good is this statistic in describing the data?

   (c) A subcommand of the `mean()` command is "trim". Use `?mean` to investigate what this subcommand does, and experiment with various values of "trim".

   (d) Calculate the median of the `INCOME` data using `median(INCOME)`. Does this give a better "feel" for the data set than the mean?

   (e) Calculate the 0.1 quantile and the third quartile for the `INCOME` data. Use, e.g., `quantile(INCOME, 0.1)`.

8. Load the Old Faithful Geyser data with

```
data(faithful)
```

Type also `?faithful` at the prompt to obtain the documentation on these data. Make available also the individual vectors `eruptions` and `waiting` in the data frame `faithful` by typing

```
attach(faithful)
```

For each of these two vectors, experiment with functions `length`, `sum`, `min`, `max`, `mean`, `median`, `sd`, and `summary`, e.g.,

```
length(eruptions)
```

(which just gives the length of the vector). If necessary, consult the documentation on any of these functions.

Use the function `hist` to show the distributions of both eruptions and waiting. Now investigate the relationship between the two variables with

```
plot(eruptions,waiting)
```

Give a detailed, but concise, description (about 200 words) of the important features of the data and of the relationship between these two variables.

9. A coin is given 10 independent tosses, each of which lands heads with probability p. Let the random variable N denote the total number of heads obtained. Then, in R, given p and any value of n, we can calculate the probability of n heads as

```
dbinom(n,10,p)
```

Set $p = 0.3$ (not a very fair coin!) and find the probabilities of various values of n in the range 0 to 10.

Now generate the entire probability function of the random variable N with

```
pf = dbinom(0:10,10,0.3)
```

Use `sum` to verify that the sum of these probabilities is 1, and plot a bar chart to show their distribution with

```
barplot(pf, names = 0:10)
```

What are the first, second, and third most likely numbers of heads? Calculate the mean and standard deviation of this distribution with

```
mn = sum(0:10 * pf)
stdev = sqrt(sum((0:10)^2 * pf) - mn^2)
```

Verify that the values obtained agree with those obtained from your own knowledge of the binomial distribution.